RASYADWA ARSYA IRNANTYANTO

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Information Technology undergraduate at Universitas Gadjah Mada with hands-on experience in IoT systems, embedded AI, and full-stack development. Experienced in building end-to-end technology solutions across hardware, firmware, cloud infrastructure, and user-facing applications, with project exposure in water infrastructure, medical wearables, and smart environmental monitoring. Strong interest in applying technology to sustainability, public health, and community-based innovation.

EDUCATION

Universitas Gadjah Mada – Yogyakarta, Indonesia Aug 2024 – Aug 2028 (Expected)
Bachelor of Information Technology | GPA: 3.39 / 4.00

- Selected for the national stage of PKM-KC 2025 (score 675/700) as Full-Stack Developer for a bronchopneumonia early detection device integrating machine learning and real-time web visualization.
- Secured national funding for PKM-KC 2026 (score 655/700) as AI Engineer for OSTOSENSE, an ostomy leakage risk monitoring wearable using edge-based machine learning.

SMAN 1 Kendari Barat – Kendari, Southeast Sulawesi Apr 2021 – Feb 2024
Senior High School, Science Major | 94.96 / 100.00

- National finalist of KIHAJAR STEM 2022, representing Southeast Sulawesi province.
- Provincial finalist of Kompetisi Sains Nasional (KSN) 2021 and 2022 in Physics, representing Southeast Sulawesi.

WORK EXPERIENCE

HydroConnect – IEEE SIGHT × PT Telkom Indonesia CSR | Gunungkidul, Indonesia Mar 2025 – Mar 2026
Vice Project Director, Power Generation & Instrumentation

- Led deployment of hybrid solar-battery system (3×440Wp panels, 3000W inverter, 2×150Ah Li-ion) powering clean water infrastructure for 379 residents in Dusun Pengos.
- Oversaw IoT instrumentation for real-time water quality monitoring (TDS, turbidity, pH, temperature, DO) via LILYGO SIM7600G; live data dashboard published on Google Play Store.
- Supported system capacity planning, projecting 69% water demand fulfillment and 78% reduction in PLN grid dependency through hybrid solar-battery integration.

PROFESSIONAL & ACADEMIC PROJECTS

OSTOSENSE – PKM-KC 2026, DTETI FT UGM 2026
AI Engineer

- Designed edge inference pipeline on ESP32-S3 using L2-regularized ordinal logistic regression for real-time ostomy leakage risk stratification (4-class: Safe / Monitor / Caution / Urgent).
- Developed ML feature extraction from LIG resistive and capacitive sensor time-series and integrated the model with Firebase, NestJS API, and React Native mobile application.

IoT Smart Waste Management – PKM FT UGM, Kalurahan Srimulyo 2026
Hardware Contributor & Stakeholder Liaison

- Contributed to hardware development with multi-sensor nodes (HC-SR04, HX711, DHT22, MQ-4, MQ-135, ESP32-CAM) communicating via LoRa with solar power supply.
- Coordinated with the student team, faculty advisor, and village officials to align hardware development with local implementation requirements.

PohonKu – Adopt-a-Tree Marketplace 2026
Frontend Developer & Project Manager

- Developed the React frontend for an adopt-a-tree marketplace integrating PostgreSQL-backed data, Midtrans payment flow, and Google OAuth authentication.
- Managed product roadmap, task delegation, and coordination between frontend and backend development workstreams.

Agent-Based Fish Schooling Simulation – Simulation Modeling Course, DTETI FT UGM 2026 2026
Sole Developer & Project Leader

- Developed a Unity/C# agent-based simulation prototype modeling emergent fish-schooling behavior under stochastic water-current disturbances.
- Implemented fish-agent behavioral rules, Box-Muller-based current disturbance generation, real-time visualization, adjustable parameters, and quantitative group-coherence analysis.

ORGANIZATIONAL EXPERIENCE

Technocorner UGM 2026 – KMTETI FT UGM Oct 2025 – Jun 2026
IoT Development Track Coordinator

- Coordinated technical evaluation of IoT competition submissions, covering embedded systems, sensor networks, connectivity, and cloud integration.
- Managed scoring rubrics, judge coordination, participant briefings, and submission review process for IoT Development Track.

FIND IT! UGM 2026 – KMTETI FT UGM Oct 2025 – May 2026
Sub-Coordinator, AI Hackathon Competition

- Coordinated competition mechanics, participant communication, and technical judging criteria for a 24-hour AI Hackathon track in collaboration with AI Connect by Telkom.

KMTETI FT UGM Jan 2025 – Dec 2025
Head of Social Affairs Division

- Led community development programs including Desa Binaan initiatives in Pengos and Kalurahan Srimulyo, coordinating field activities, volunteer involvement, and stakeholder communication within a 250+ member student organization.

NIGHTLOGIN BSO – DTETI FT UGM Jan 2025 – Present
Lead, Human Resources Division

- Coordinated HR programs, member development activities, and internal-external opportunity sharing for NIGHTLOGIN BSO members.

ENFORIAN 2024 – DTETI FT UGM Sep 2024 – Dec 2024
Vice Head of Production

- Managed technical production and logistics coordination for a theatrical event involving 250+ students across multiple divisions.

ACHIEVEMENTS

National Finalist – KIHAJAR STEM 2022 (representing Southeast Sulawesi) 2022

Provincial finalist – Kompetisi Sains Nasional (KSN) in Physics (representing Southeast Sulawesi) 2021 & 2022

PKM-KC UGM Internal Selection Score: 675/700 – Bronchopneumonia Early Detection Device 2025

PKM-KC UGM National Funding Stage Score: 655/700 – OSTOSENSE, a cyber-physical medical wearable for ostomy leakage risk stratification using edge inference on ESP32-S3 with ordinal logistic regression. 2026

CERTIFICATIONS

Deep Learning in Python – DataCamp May 2026
18-hour course covering neural networks, Keras, and deep learning model optimization.

CCNA: Introduction to Networks – Cisco Networking Academy June 2026
Foundational networking concepts including IP addressing, routing, switching, and network protocols.

SKILLS

Programming: Python, JavaScript, C, C+, C++

Frameworks: React, Next.js, NestJS, React Native

IoT & Embedded: ESP32, ESP32-S3, STM32, LoRa, MQTT, Firebase

AI / ML: Ordinal Logistic Regression, Decision Trees, Naive Bayes, DBSCAN, Deep Learning (Keras), NumPy, Pandas

Cloud & Database: Firebase (GCP), PostgreSQL

Networking: IP Addressing, Routing & Switching, Network Protocols (CCNA)

Tools: KiCad, STM32CubeIDE, Git, Figma, Unity

Languages: Indonesian (Native), English (Proficient)